

IGNAT'YEVO, Russian Federation

WMO: 313289

Lat: 50.425N

Lon: 127.412E

Elev: 638

StdP: 14.36

Time Zone: 9.00 (E09)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.0	(c) -22.3	(d) -32.9	(e) 0.9	(f) -25.2	(g) -29.0	(h) 1.1	(i) -21.1	(j) 18.3	(k) -0.2	(l) 16.3	(m) -0.5	(n) 2.6	(o) 250	(p) 0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	86.4	69.7	84.0	68.8	80.8	67.7	74.4	81.5	72.7	79.4	71.1	77.3	8.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	120.9	77.6	70.2	113.7	75.5	68.4	106.9	74.0	38.3	81.5	36.7	78.8	35.3	78.0	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	17.3	15.0	-33.9	91.9	3.2	2.5	-36.2	93.6	-38.1	95.1	-39.9	96.5	-42.2	98.3
			-33.6	77.4	3.3	1.8	-36.0	78.7	-37.9	79.7	-39.8	80.7	-42.2	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	-5.9	1.4	19.9	40.0	54.3	64.6	70.8	67.3	54.8	36.6	13.3	-3.8
	DBStd	28.73	8.63	10.21	11.94	9.36	6.98	6.10	4.46	5.29	8.01	8.98	12.39	8.31
	HDD50	7622	1733	1361	934	325	33	1	0	0	42	424	1101	1668
	HDD65	11472	2198	1781	1399	751	337	82	7	35	316	882	1551	2133
	CDD50	2005	0	0	0	25	168	438	645	535	186	8	0	0
	CDD65	380	0	0	0	2	6	69	187	106	10	0	0	0
	CDH74	3293	0	0	0	41	201	815	1352	767	117	0	0	0
	CDH80	810	0	0	0	11	44	257	342	141	15	0	0	0
(14) Wind	WSAvg	6.3	4.7	5.2	7.0	8.8	8.0	6.1	5.9	5.5	6.6	7.0	6.0	5.1
(15) Precipitation	PrecAvg	21.6	0.2	0.2	0.4	0.9	2.1	3.3	5.7	4.2	2.6	1.0	0.5	0.4
	PrecMax	31.1	1.2	1.2	1.1	3.5	5.0	8.4	12.7	9.7	5.3	3.4	1.0	1.2
	PrecMin	14.2	0.0	0.0	0.0	0.0	0.4	0.4	0.5	1.0	0.5	0.1	0.0	0.1
	PrecStd	4.4	0.2	0.2	0.4	0.7	1.3	1.9	2.8	2.2	1.5	0.8	0.3	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	34.2	53.9	79.2	85.7	91.1	89.9	87.6	82.2	68.3	51.5	20.8
		MCWB	13.8	26.8	41.4	53.6	59.1	67.7	73.7	73.7	65.8	51.1	41.2	18.1
	2%	DB	12.4	28.1	47.9	69.7	80.2	86.4	86.4	83.8	77.2	62.2	43.1	16.2
		MCWB	9.8	23.2	37.5	49.6	57.5	68.0	72.6	70.5	61.5	48.6	36.5	14.5
	5%	DB	10.0	22.9	42.9	62.4	75.1	82.5	84.0	80.8	73.2	55.8	37.1	12.4
		MCWB	8.1	20.0	33.8	46.1	55.1	65.6	71.4	69.1	61.1	45.3	32.2	10.8
	10%	DB	6.7	17.7	38.8	57.1	70.1	78.7	80.8	78.7	68.3	51.6	30.5	8.5
		MCWB	5.1	15.3	31.5	42.6	53.5	64.7	69.7	67.8	57.9	42.0	28.1	7.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	27.8	41.2	55.6	63.1	72.8	76.5	77.1	68.7	55.9	40.9	18.4
		MCDB	16.9	33.8	53.0	77.2	77.7	81.5	83.9	84.2	73.8	64.1	47.5	20.2
	2%	WB	10.0	23.5	37.5	50.5	60.0	70.4	75.0	73.9	65.5	50.3	37.3	14.6
		MCDB	12.4	27.5	46.2	66.0	72.2	80.3	82.3	80.0	71.0	56.7	45.7	16.3
	5%	WB	7.5	19.9	34.0	47.0	57.9	68.6	73.6	71.7	63.0	46.6	31.9	11.0
		MCDB	9.4	22.8	41.9	60.3	70.4	78.4	80.9	77.4	69.6	53.5	36.6	12.5
	10%	WB	5.0	15.3	31.3	43.6	55.6	66.7	72.2	69.9	60.8	43.1	28.0	7.0
		MCDB	6.3	17.5	38.1	55.0	66.9	76.0	78.6	76.1	67.3	50.3	31.1	8.2
(35) Mean Daily Temperature Range	MDBR		16.7	21.3	21.3	22.8	23.7	21.4	18.8	19.4	21.8	19.7	16.5	15.4
	5% DB	MCDBR	16.8	22.1	22.8	30.9	32.4	29.1	22.2	22.9	26.4	27.2	21.2	17.6
		MCWBR	14.6	18.1	14.3	17.0	14.2	13.2	10.6	12.3	13.2	16.5	15.2	15.6
	5% WB	MCDBR	16.4	21.2	22.4	28.0	26.3	23.1	18.2	18.9	20.6	23.0	19.7	17.0
		MCWBR	14.3	17.4	14.6	16.0	13.8	12.2	10.5	11.3	11.6	16.1	15.0	15.1
(40) Clear-Sky Solar Irradiance	taub		0.269	0.300	0.372	0.509	0.568	0.498	0.479	0.427	0.390	0.383	0.315	0.276
	taud		2.165	2.095	1.954	1.782	1.726	2.012	2.118	2.252	2.274	2.128	2.155	2.120
	Ebn at Noon		238	259	256	231	224	244	247	255	252	223	213	212
	Edh at Noon		24	34	47	63	70	53	47	39	34	33	24	22
(44) All-Sky Solar Radiation	RadAvg		525	907	1399	1590	1727	1862	1724	1528	1223	813	522	409
	RadStd		17	30	89	167	141	236	144	104	103	48	29	22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page